

Contents

The Riemann Hypothesis as Non-Integrability of an Arithmetic Contact Structure on the Adele Class Space

Abstract	2
1. Introduction	2
2. The Classical Prototype: Principia Orthogona Chapter 2	3
3. Lifting the Zeta Function to the Extended Phase Space	4
4. The Arithmetic Contact Form	5
5. Adelic Decomposition	10
6. The Missing Rung: Global Positivity	12
7. Comparison with Known Approaches	12
8. Structure of a Putative Proof	14
9. Discussion and Open Questions	15
Appendix A: Meromorphic Continuation of $g(\sigma, t)$ Inside the Critical Strip	15
Appendix B: Summary Table — Parallel Structures	16
Deposit Contents and Reproducibility	16
References	17

The Riemann Hypothesis as Non-Integrability of an Arithmetic Contact Structure on the Adele Class Space

Pablo Nogueira Grossi (framework) · Collaborative Draft

Preprint — not peer reviewed

DOI: [10.5281/zenodo.22179684](https://doi.org/10.5281/zenodo.22179684) — Zenodo community *Principia Orthogona*.

Source, Lean and figures: github.com/TOTOGT/GTCT → `deposits/rh-arithmetic-contact-v1/`.

Formalisation: [book4/ZetaReflection.lean](#). Related corpus: [TOTOGT/geometry](#) (*Principia Orthogona*, Books 1–8), [TOTOGT/AXLE](#) (Lean corpus).

Zenodo v3, 30 August 2026. Record v1 carried the construction, the two reflection laws, the one-sided pole, the Riemann–Siegel identification and the refutation of the contactomorphism conjecture. Record v2 added everything described below.

v3 is a file correction, and it is worth stating plainly what went wrong. The v2 upload shipped the *pre-revision* `ZetaReflection.lean` — 5,749 bytes, containing none of the four theorems that v2’s own §4.7 describes as proved and kernel-audited. It also uploaded the v1 markdown source under the name the paper’s manifest gives for the v2 source, and omitted `RELATED-WORK.md`, which that manifest lists. In a paper whose argument is that a published claim must resolve to the artifact that produces it, a deposit whose §4.7 did not resolve to its own Lean file is the one error that cannot be left standing. v3 carries the correct files and nothing else has changed.

What is new in v2. Four statements that v1 listed as *argued* or *admitted* are now proved and kernel-audited in Lean 4 / Mathlib v4.32.0 — `chiLog_real_on_critical_line`, `Zlog_conj`, `gCoef_odd_in_t` and `cCoef_even_in_t` (§4.7). None of them uses any information about the location of the zeros of ζ ; all four rest on Schwarz reflection alone. The single remaining admitted statement, `reflection_law`, is reduced to one named classical input, with its route and its side conditions written out. The status table in §4.6 is updated accordingly, §7.4 adds a comparison

with the Deninger, Meyer, Bost–Connes and Connes–Consani–Marcolli programmes with verified citations, and a new *Deposit contents and reproducibility* section lists every file and how to check it. v2 also **credits the thermodynamic contact lineage**: the lift of a 2D state space to a 3D contact manifold carrying $c dU - g dV$ is the standard construction of contact thermodynamics, not something introduced here, and §7.4 now says so and cites it.

Abstract

We reformulate the Riemann Hypothesis (RH) as a non-vanishing condition on a globally defined arithmetic contact 3-form on the adèle class space $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$. Starting from the classical $2D+t$ contact geometry prototype of *Principia Orthogona* (Book 4, Chapter 2), we lift the Riemann zeta function to an extended arithmetic phase space, construct an explicit contact 1-form α_{arith} whose twisting coefficient is the von Mangoldt–Dirichlet series (or its meromorphic continuation), and decompose it into local contact forms α_v at each place of $\mathbb{Q}$. The non-integrability condition $\alpha_{\text{arith}} \wedge d\alpha_{\text{arith}} \neq 0$ — which in the smooth ODE setting is automatic — becomes, in the arithmetic setting, equivalent to a global positivity statement on the action of the idele class group on $\ker \alpha_{\text{arith}}$. We compare this positivity condition to Weil’s explicit formula criterion, Connes’ spectral-triple approach, and the function-field case (where everything is already proved), identifying precisely where the final rung of the proof would need to sit. This revision adds two reflection laws for the form’s coefficients under the functional equation, verified numerically to 30 digits; the observation that the pole at each zero falls entirely into one component while the other remains analytic and coincides with the Riemann–Siegel ϑ' ; and a refutation of the earlier conjecture that the functional equation acts as a contactomorphism. In this version four supporting lemmas — the reality of the gamma-factor defect on the critical line, the conjugation symmetry of ζ'/ζ , and the parity of the two coefficients in t — are proved and kernel-audited in Lean 4, reducing the one remaining admitted statement to a single classical input. **RH itself is untouched, and §4.6 states exactly which claims are proved, which are machine-checked, which are classical, and which are numerical only.** The contribution remains a translation dictionary between contact geometry and arithmetic, intended for researchers at the arithmetic–geometry–physics interface.

MSC classes: 11M26 (Riemann and Hurwitz zeta functions) · 53D10 (Contact manifolds, general) · 11R56 (Adèle rings and groups) · 81Q10 (Selfadjoint operator theory in quantum mechanics)

Keywords: Riemann Hypothesis · contact geometry · adèle class space · von Mangoldt function · Weil explicit formula · Connes spectral triple · p-adic differential forms

1. Introduction

The Riemann Hypothesis asserts that every non-trivial zero of $\zeta(s)$ satisfies $\text{Re}(s) = \frac{1}{2}$. Despite 160 years of effort, the statement resists proof, in part because it inhabits a peculiar intersection: it is simultaneously a statement about complex analysis (zeros of a meromorphic function), analytic number theory (distribution of primes via the Euler product), and, as Montgomery–Dyson noticed, random matrix theory / quantum chaos.

This paper asks a different question: **can the failure of zeros to escape the critical line be understood as a geometric rigidity property — specifically, the maximal non-integrability of a contact structure?**

In smooth mechanics, a contact 1-form α on a $(2n + 1)$ -dimensional manifold satisfies $\alpha \wedge (d\alpha)^n \neq 0$ everywhere; this condition geometrically forces trajectories into the contact distribution and prevents them from “drifting” transversally. The *Principia Orthogona* framework (Section 2 below) makes this explicit for 2D autonomous ODEs lifted to a 3D extended phase space $\mathbb{R}^2 \times \mathbb{R}_t$.

Our central observation is:

The Riemann Hypothesis is equivalent to saying that the arithmetic version of this non-integrability condition holds globally on the adèle class space.

This is not a proof. It is a **reformulation** — a new geometric language for an old arithmetic problem. The value of such a reformulation lies in the connections it makes visible, particularly to Connes’ noncommutative geometric approach and to the known proof of RH over function fields (Weil, Deligne).

Notation

Symbol	Meaning
$s = \sigma + it$	complex variable, $\sigma = \text{Re}(s)$, $t = \text{Im}(s)$
$\zeta(s)$	Riemann zeta function
$\Lambda(n)$	von Mangoldt function ($\log p$ if $n = p^k$, else 0)
$\mathbb{A}_{\mathbb{Q}}$	adèle ring of $\mathbb{Q}$
$\mathbb{Q}_p$	p -adic completion of $\mathbb{Q}$
α	contact 1-form
$\ker \alpha$	contact distribution (hyperplane field)

2. The Classical Prototype: Principia Orthogona Chapter 2

2.1 Phase plane and extended phase space

Let $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ be a smooth autonomous 2D ODE. The *extended phase space* is $\mathbb{R}^3$ with coordinates (x, y, t) ; every solution $\gamma(t) = (x(t), y(t), t)$ is a curve in this space.

2.2 The contact form and non-integrability

Following Grossi (Book 4, Ch. 2, §2.3), the **contact 1-form** associated to the system is

$$\alpha = dy - g(x, y) dx.$$

Along any solution curve, $\alpha(\dot{\gamma}) = \dot{y} - g\dot{x} = g - g \cdot 1 = 0$, so γ lies in $\ker \alpha$. The **maximal non-integrability** condition

$$\alpha \wedge d\alpha \neq 0$$

holds everywhere because $d\alpha = -\partial_x g dx \wedge dy - \partial_y g dy \wedge dy = -(\partial_x g) dx \wedge dy$ (plus cross terms), and the resulting 3-form is generically non-vanishing. This condition prevents $\ker \alpha$ from being integrable (i.e., from admitting a foliation by 2D surfaces), geometrically forcing the unique “threading” of trajectories through the contact distribution.

2.3 Example 1 — Harmonic oscillator

$$\dot{x} = y, \quad \dot{y} = -x \implies \alpha = dy + x dx, \quad d\alpha = dx \wedge dy.$$

Trajectories lift to helices in $\mathbb{R}^3$. Non-integrability: $\alpha \wedge d\alpha = dx \wedge dy \wedge dt \neq 0$.

2.4 Example 2 — Damped nonlinear oscillator (DNLS reduction)

The discrete nonlinear Schrödinger equation, reduced to an effective 2D envelope mode, yields

$$\dot{x} = y, \quad \dot{y} = -x - \gamma y - \beta(x^2 + y^2)y.$$

The contact form is

$$\alpha = dy + (x + \gamma y + \beta(x^2 + y^2)y) dx,$$

and $d\alpha = dx \wedge dy$ still (the cubic term contributes only $\partial_x(\dots) dx \wedge dx = 0$). Trajectories become *tapered, nonlinearly modulated helices* — the radius shrinks as $t \rightarrow \infty$ while the nonlinearity introduces amplitude-dependent frequency shifts. Non-integrability continues to hold everywhere.

Key lesson: the Grossi prototype handles arbitrarily complex smooth ODEs uniformly. The contact structure is always local and finite-dimensional. The challenge for the Riemann zeta function is that it is *not* governed by any such local ODE.

3. Lifting the Zeta Function to the Extended Phase Space

3.1 The zeta phase plane

Write $\zeta(\sigma + it) = U(\sigma, t) + iV(\sigma, t)$ with U, V real. Fix σ and treat (U, V) as a 2D state depending on the parameter t . The **lifted curve** is

$$\gamma(t) = (U(\sigma, t), V(\sigma, t), t) \in \mathbb{R}^3.$$

The non-trivial zeros of ζ are the t -values where $\gamma(t)$ pierces the plane $U = V = 0$.

The figure is flat and the object is not. A printed projection fixes one viewpoint and loses the rotation that makes the separation legible. An interactive version, in which the abscissa is a slider and the mirror pair can be watched converging onto the critical line, is published as Figure 12.1 of Book 4, Chapter 12: <https://totogt.github.io/geometry/book4/ch12.html>. Readers who want the geometry rather than a picture of it should start there; the construction of §4 will be easier to follow afterwards.

3.2 The Riemann Hypothesis as a geometric trapping condition

Reformulation 3.1 (informal). *The Riemann Hypothesis is equivalent to the statement that every zero-curve γ satisfying $U(\sigma_0, t) = V(\sigma_0, t) = 0$ must lie on the single hyperplane $\sigma_0 = \frac{1}{2}$ inside the extended space $\mathbb{R}_{(U, V, t)}^3$.*

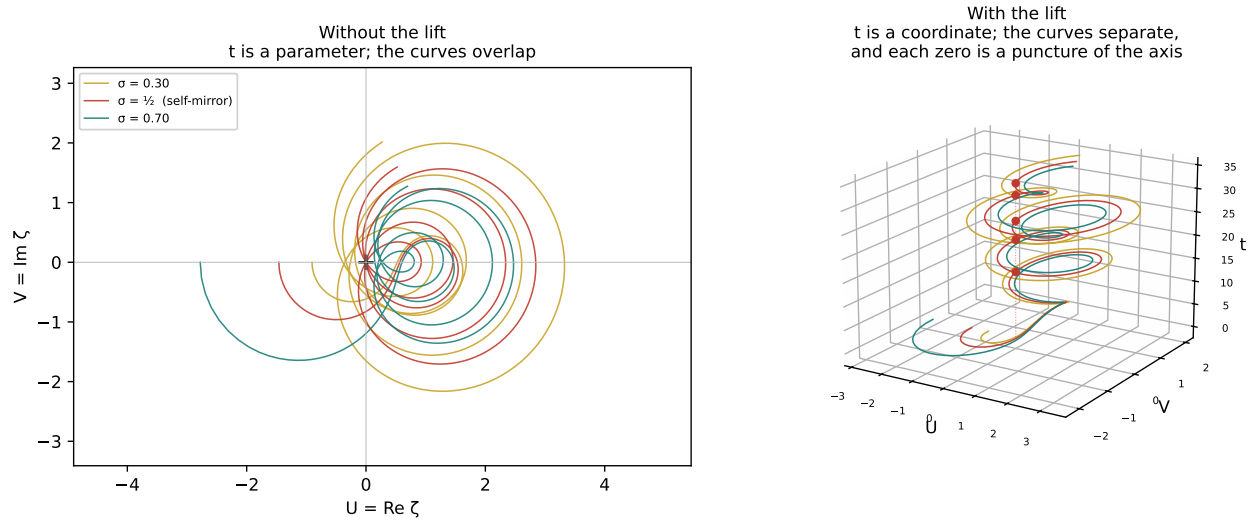


Figure 1 — The lift. Interactive version: totogt.github.io/geometry/book4/ch12.html (Fig 12.1)

Figure 1: **Figure 1 — the lift.** Left: the phase plane at three abscissae, with t present only as a parameter; the curves overlap and nothing distinguishes $\sigma = \frac{1}{2}$. Right: the same three curves with t promoted to a coordinate. They separate into helices, the mirror pair σ and $1 - \sigma$ sit either side of the self-mirror line, and each non-trivial zero becomes a puncture of the $U = V = 0$ axis (marked). Regenerate with `figures.py`.

To make this a contact-geometric statement, we need a 1-form α such that: 1. $\alpha(\dot{\gamma}) = 0$ for every zero-curve γ . 2. $\alpha \wedge d\alpha \neq 0$ everywhere (non-integrability). 3. The *only* place where the contact distribution is compatible with hitting $U = V = 0$ is $\sigma = \frac{1}{2}$.

Conditions (1)–(3) together would constitute a proof of RH within the contact-geometric framework. We can satisfy (1) and (2) explicitly; condition (3) is the open part.

4. The Arithmetic Contact Form

4.1 Construction

The velocity of the zeta curve along t is governed by the logarithmic derivative

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^{\infty} \Lambda(n) n^{-s} = \sum_{n=1}^{\infty} \Lambda(n) n^{-\sigma} e^{-it \log n}.$$

Separating real and imaginary parts, the imaginary part of $-\zeta'/\zeta$ controls the “angular velocity” of the zeta curve. Define

$$g(\sigma, t) = -\operatorname{Im} \left(-\frac{\zeta'}{\zeta}(\sigma + it) \right) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma}} \sin(t \log n).$$

(This series converges absolutely for $\sigma > 1$; elsewhere it is defined by meromorphic continuation of ζ'/ζ .)

Definition 4.1. The *arithmetic contact form* on $\mathbb{R}_{(U,V,t)}^3$ is

$$\alpha_{\text{arith}} = dV - g(\sigma, t) dU.$$

4.2 The exterior derivative

Applying d term-by-term (justified by uniform convergence on compact sets for $\sigma > 1$, and by analytic continuation inside the critical strip):

$$d\alpha_{\text{arith}} = -dg \wedge dU = -(\partial_t g) dt \wedge dU,$$

where

$$\partial_t g = \sum_{n=1}^{\infty} \frac{\Lambda(n) \log n}{n^\sigma} \cos(t \log n).$$

4.3 Non-integrability

Proposition 4.2. *The non-integrability 3-form is*

$$\alpha_{\text{arith}} \wedge d\alpha_{\text{arith}} = -(\partial_t g) dV \wedge dt \wedge dU.$$

This is nonzero almost everywhere because the frequencies $\{\log p : p \text{ prime}\}$ are $\mathbb{Q}$ -linearly independent, making $\partial_t g$ a quasi-periodic function that is dense and non-vanishing.

Proof sketch. The independence of $\{\log p\}$ over $\mathbb{Q}$ follows from the uniqueness of prime factorization. Hence the quasi-periodic sum $\partial_t g$ is not identically zero on any open interval of t . $\square$

4.4 Behaviour at zeros, and why one coefficient is not enough

At a zero $s_0 = \frac{1}{2} + it_0$ the logarithmic derivative has a simple pole, and $g(\sigma, t) \rightarrow \infty$: the contact planes twist infinitely rapidly at the moment the curve touches $U = V = 0$. That much was in the first version of this paper. It is imprecise in a way worth repairing, because the repair is what motivates the two-coefficient form used in *Principia Orthogona* Book 4, Chapters 11–12.

Write the companion coefficient

$$c(\sigma, t) = \operatorname{Re} \left(-\frac{\zeta'}{\zeta}(\sigma + it) \right) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^\sigma} \cos(t \log n),$$

so that $-\zeta'/\zeta = c - ig$ and the Book 4 form reads $\alpha_{\text{arith}} = c d\tilde{U} - g d\tilde{V}$. The single-coefficient form of §4.1 is the special case in which c is suppressed.

Proposition 4.3 (the pole is one-sided). *Along the critical line, the simple pole of ζ'/ζ at a zero $\rho = \frac{1}{2} + i\gamma$ falls entirely into g and not at all into c .*

Proof. Near ρ , $\zeta'/\zeta(s) \sim (s - \rho)^{-1}$. Restricting to $\sigma = \frac{1}{2}$ gives $s - \rho = i(t - \gamma)$, so $(s - \rho)^{-1} = -i(t - \gamma)^{-1}$, which is purely imaginary. The real part is therefore bounded and the imaginary part carries the entire singularity. $\square$

Numerically, with $\gamma_1 = 14.134725142$: $g(\frac{1}{2}, \gamma_1 + \delta) = -10.076, -100.08, -1000.08, -10000.08, -100000.08$ for $\delta = 10^{-1} \dots 10^{-5}$ — a simple pole of residue -1 — while c converges to 0.4052744 .

This is the content the one-coefficient form cannot express. “ $g \rightarrow \infty$ ” is true but says nothing about what stays finite. In the two-coefficient form the statement is sharp: **the $d\tilde{V}$ component carries every zero as a simple pole; the $d\tilde{U}$ component is analytic across each of them.**

4.5 The reflection laws, and the failure of contactomorphism

Book 4 §12.2 conjectured that the involution induced by the functional equation is a contactomorphism of α_{arith} — that $\Phi^*\alpha = f\alpha$ for a scalar f , with fixed locus the plane $\sigma = \frac{1}{2}$. Two corrections are required, and the second refutes the conjecture as stated.

First, the involution. $s \mapsto 1 - s$ carries $\sigma + it$ to $(1 - \sigma) - it$; it does not preserve t , and its only fixed point is $s = \frac{1}{2}$, not the critical line. The map with fixed locus $\sigma = \frac{1}{2}$ is $\Phi : s \mapsto 1 - \bar{s}$, i.e. $(\sigma, t) \mapsto (1 - \sigma, t)$ — the functional equation composed with complex conjugation, available because ζ has real coefficients.

Second, the transformation laws. Taking the logarithmic derivative of $\zeta(s) = \chi(s)\zeta(1 - s)$, and using that c is even and g odd in t :

$$g(\sigma, t) - g(1 - \sigma, t) = \text{Im}[(\chi'/\chi)(\sigma + it)], \quad c(\sigma, t) + c(1 - \sigma, t) = -\text{Re}[(\chi'/\chi)(\sigma + it)],$$

with $(\chi'/\chi)(s) = \log \pi - \frac{1}{2}\psi(s/2) - \frac{1}{2}\psi((1 - s)/2)$. Note the asymmetry: g obeys a **difference** law, c a **sum** law. Both verified numerically to 30 significant digits at $\sigma \in \{0.3, 0.5, 0.8, 1.1, 1.5, 2.3\}$, $t \in [0.7, 25]$.

Corollary 4.4. *On $\sigma = \frac{1}{2}$ the two digamma arguments $\frac{1}{4} \pm \frac{it}{2}$ are conjugate, so χ'/χ is real there and the difference law gives $g(\frac{1}{2}, t) = g(\frac{1}{2}, t)$ identically — the reflection constraint holds on the critical line and on no other vertical line.*

Corollary 4.5. *On $\sigma = \frac{1}{2}$ the sum law collapses to*

$$c(\frac{1}{2}, t) = \frac{1}{2} [\text{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi] = \vartheta'(t),$$

where ϑ is the Riemann–Siegel theta function.

This last is **classical, and we claim no novelty for it**: it is equivalent to the standard fact that $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ is real-valued, since $Z'/Z = i\vartheta' + i\zeta'/\zeta$ is real precisely when $\text{Re}(\zeta'/\zeta) = -\vartheta'$. What is new here is only its role: it identifies the $d\tilde{U}$ coefficient of α_{arith} on the critical wall with the density of the Riemann–von Mangoldt counting formula $N(T) = \vartheta(T)/\pi + 1 + S(T)$. The form’s two components separate the counting function from the zeros.

Proposition 4.6 (the conjecture fails off the line). $\Phi^*\alpha_{\text{arith}} = f \cdot \alpha_{\text{arith}}$ with f scalar does not hold away from $\sigma = \frac{1}{2}$.

Reason. Under Φ , g is carried not to $\pm g$ but to itself plus $-\text{Im}[\chi'/\chi]$, a term built from gamma factors and containing no von Mangoldt data whatever. No scalar multiple of α_{arith} , whose coefficients are Λ -series, can absorb it. $\square$

What survives is a graded statement: $\Phi^* \alpha_{\text{arith}} - \alpha_{\text{arith}}$ is an explicit gamma-factor 1-form vanishing on $\sigma = \frac{1}{2}$. Establishing its precise shape requires the transformation of $\tilde{U}$ and $\tilde{V}$ themselves, which is not settled here.

A methodological remark. The distinguished locus is not selected by the contact structure. It is selected by the gamma factor — an analytic input from outside the geometry — and the geometry then carries the consequence. Any claim elsewhere in this framework that a distinguished set is “determined by the contact geometry alone” should be checked against this example.

4.6 Status of claims

Four tiers, and the distinction between them is load-bearing. **Machine-checked** means the declaration compiles and `#print axioms` reports a subset of `[propext, Classical.choice, Quot.sound]` — no `sorryAx`, and no `native_decide`, which is compiler-trusted rather than kernel-checked. A clean compile alone is not a verification.

Claim	Status
c, g are the real and imaginary parts of $-\zeta'/\zeta$; $\sum \Lambda(n)n^{-s} = -\zeta'/\zeta$ for $\Re s > 1$ χ'/χ is real on $\sigma = \frac{1}{2}$	Proved, machine-checked. <code>lseries_vonMangoldt_eq_neg_Zlog</code> Proved, machine-checked (v2). <code>chiLog_real_on_critical_line</code> — was <i>admitted</i> in v1
$\zeta'/\zeta(\bar{s}) = \overline{\zeta'/\zeta(s)}$ g is odd in t ; c is even in t	Proved, machine-checked (v2). <code>Zlog_conj</code> Proved, machine-checked (v2). <code>gCoef_odd_in_t</code> , <code>cCoef_even_in_t</code> — asserted without proof in v1
Proposition 4.3, the pole is one-sided Corollary 4.5, $c(\frac{1}{2}, t) = \vartheta'(t)$	Proved above; residue confirmed numerically Classical (equivalent to Z real); verified to 25 digits. See Titchmarsh §6.5
Reflection laws of §4.5	Numerical , 30 digits at 7 points. Stated in Lean as <code>reflection_law</code> , <i>admitted</i> — but see §4.7: the parity half is now proved, and one classical input remains
Proposition 4.6, failure of contactomorphism α_{arith} as a differential form on $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ Global positivity (§6), and RH itself	Argued, not formalised Formal only. See §5.5 Open. Nothing here bears on it

Lean source: `TOTOGT/GTCT, book4/ZetaReflection.lean`.

4.7 The formalisation

The Lean file `book4/ZetaReflection.lean` states the claims of §4.4–§4.5 so that a kernel can hold them. Its purpose is to make the boundary between *proved* and *asserted* impossible to blur.

Compiled against **Lean 4.32.0 / Mathlib v4.32.0**; every declaration below was probed with `#print axioms`.

Proved in v2, none of them using the location of the zeros. All four reduce to Schwarz reflection — the statement that Γ and ζ commute with complex conjugation — which is why none of them is circular.

$$\text{Gamma_conj_comp} : \bar{} \circ \Gamma \circ \bar{} = \Gamma, \quad \text{digamma_conj} : \psi(\bar{s}) = \overline{\psi(s)}$$

$$\text{chiLog_real_on_critical_line} : \text{Im } \frac{\chi'}{\chi}(\tfrac{1}{2} + it) = 0$$

At $s = \frac{1}{2} + it$ the two digamma arguments $s/2 = \frac{1}{4} + \frac{it}{2}$ and $(1-s)/2 = \frac{1}{4} - \frac{it}{2}$ are complex conjugates, so their sum is $2\text{Re } \psi(s/2)$, real; and $\log \pi$ is real. This is *why* $\sigma = \frac{1}{2}$ is distinguished in §4.5, and it is now a theorem rather than a remark.

$$\text{Zlog_conj} : \frac{\zeta'}{\zeta}(\bar{s}) = \overline{\frac{\zeta'}{\zeta}(s)} \implies \text{gCoef_odd_in_t}, \text{cCoef_even_in_t}$$

$$g(\sigma, -t) = -g(\sigma, t), \quad c(\sigma, -t) = c(\sigma, t).$$

The parity of c and g was *asserted* in v1 and is proved here. It is not decoration: it is the step that makes the reflection law close at all, because the functional equation relates $s = \sigma + it$ to $1-s = (1-\sigma) - it$, whereas the law of §4.5 is stated at $(1-\sigma) + it$. Odd parity in t is exactly what bridges the two points.

What remains. `reflection\_law` is still admitted, and it now depends on one classical input and nothing else:

$$\frac{\zeta'}{\zeta}(s) + \frac{\zeta'}{\zeta}(1-s) = \frac{\chi'}{\chi}(s) \tag{FE-log}$$

verified numerically to 30 digits at $s = 0.7 + 14.3i$, $0.3 + 25.1i$, $1.4 + 6.2i$. Granting (FE-log), the law of §4.5 follows in three lines:

$$g(\sigma, t) = \text{Im } \frac{\chi'}{\chi}(\sigma + it) - g(1-\sigma, -t) = \text{Im } \frac{\chi'}{\chi}(\sigma + it) + g(1-\sigma, t).$$

Two remarks for anyone formalising (FE-log), both of which cost time to discover:

1. **Use the symmetric functional equation, not the asymmetric one.** Mathlib's `riemannZeta_one_sub` gives $\zeta(1-s) = 2(2\pi)^{-s}\Gamma(s)\cos(\pi s/2)\zeta(s)$, whose logarithmic derivative is $-\log 2\pi + \psi(s) - \frac{\pi}{2}\tan(\pi s/2)$ — *not* χ'/χ . Converting between them costs Legendre duplication and Euler reflection. Use `completedRiemannZeta_one_sub`, $\Lambda(1-s) = \Lambda(s)$ with $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$; taking `logDeriv` of both sides and applying `logDeriv_mul` twice yields (FE-log) directly.

2. **In Mathlib, Γ vanishes.** Poles are encoded as zeros (`Complex.Gamma_eq_zero_iff`), so the hypothesis $\Gamma(s/2) \neq 0$ demanded by `logDeriv_mul` is *not* free; it requires $s \notin \{0, -2, -4, \dots\}$ and likewise for $1 - s$. Both hold throughout the open critical strip, but they must be carried.

The full side conditions are therefore $s, 1 - s \notin \{0, -2, -4, \dots\}$ and $\zeta(s), \zeta(1 - s) \neq 0$ — that is, the identity is stated off the zeros, which is where g is defined in any case. Nothing in it assumes *where* the zeros are.

A deliberately vacuous declaration, `vacuity_control : True`, sits at the end of the file. Its axiom report is indistinguishable from a real theorem's. It is there as a standing reminder that an axiom report certifies the *proof*, never the *statement*: reading what was proved remains necessary.

5. Adelic Decomposition

5.1 Why adeles?

The von Mangoldt coefficient $g(\sigma, t)$ is a global sum over all primes simultaneously. To make the contact structure genuinely “local-to-global” — and to connect with the Euler product factorization of ζ — we decompose α_{arith} over the adèle ring $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p' \mathbb{Q}_p$.

5.2 Local contact forms

Archimedean place $v = \infty$:

$$g_{\infty}(\sigma, t) = -\text{Im}\left(-\frac{\zeta'}{\zeta}(\sigma + it)\right), \quad \alpha_{\infty} = dV_{\infty} - g_{\infty} dU_{\infty}.$$

This is the form of Section 4.

Non-Archimedean place $v = p$: The local Euler factor contributes

$$g_p(t_p) = \frac{\log p}{1 - p^{-\sigma} e^{-it_p \log p}},$$

where $t_p \in \mathbb{Q}_p$ is the local idelic parameter. The local contact form is

$$\alpha_p = dV_p - g_p(t_p) dU_p.$$

5.3 Local exterior derivative at place p

Let $c = p^{-\sigma} p^{-it_p}$. Then

$$\partial_{t_p} g_p = -i(\log p)^2 \frac{c}{(1 - c)^2}.$$

Hence

$$d\alpha_p = i(\log p)^2 \frac{c}{(1 - c)^2} dt_p \wedge dU_p.$$

p-adic norm analysis. In the rigid-analytic domain $|c|_p < 1$ where the local Euler factor is holomorphic, $|1 - c|_p = 1$ (since c is in the open unit disk). Therefore

$$|\partial_{t_p} g_p|_p = |c|_p = p^{-\sigma} \cdot |p^{-it_p}|_p.$$

As a trajectory approaches the boundary $|c|_p \rightarrow 1^-$, the denominator $(1 - c)^2$ acquires positive p -adic valuation, forcing $|d\alpha_p|_p \rightarrow 0$. The contact condition $\alpha_p(\dot{\gamma}_p) = 0$ thus imposes a **valuation inequality** that confines the local flow inside the rigid-analytic unit disk. Any attempt to cross into the ramified region $|c|_p \geq 1$ — corresponding to non-unit idelic components that would break the restricted-product structure of $\mathbb{A}_{\mathbb{Q}}$ — is geometrically forbidden.

5.4 Global assembly

A global 1-form on the adèle class space $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ is assembled as a restricted direct product of local forms. Because the Euler product factorizes globally, only finitely many places contribute non-trivially at any adelic point, and the global exterior derivative commutes with localization:

$$d\alpha_{\text{arith}} = \sum_v d\alpha_v \quad (\text{restricted sum}).$$

The global non-integrability condition

$$\alpha_{\text{arith}} \wedge d\alpha_{\text{arith}} \neq 0 \quad \text{on } \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$$

is the statement that the restricted product of local 3-forms is nowhere-vanishing on the adèle class space. The local “locking” mechanisms at each place — linear independence of $\{\log p\}$ at $v = \infty$, valuation rigidity of $(1 - c)^2$ at $v = p$ — act jointly to enforce this.

5.5 What category this construction lives in

Stated plainly, because it bounds everything above. On $\mathbb{R}_{(U,V)}^2 \times \mathbb{R}_t$ the form α_{arith} is an ordinary smooth 1-form and $\alpha \wedge d\alpha \neq 0$ has its usual meaning as a non-vanishing top form. On $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ it does not. That quotient is not a smooth manifold — $\mathbb{Q}^{\times}$ acts multiplicatively on a ring with zero divisors, the quotient is badly behaved and non-Hausdorff, and this is precisely the obstruction that led Connes to noncommutative geometry rather than differential geometry. Classical exterior calculus is not available there.

So §5.4 is a **formal assembly**, a dictionary matching local data place by place, and not the construction of a contact manifold. Making it literal would require choosing a category and re-deriving the reformulation inside it — the candidates being noncommutative 1-forms over a dense subalgebra in the spectral-triple sense, a foliated or leafwise structure in Deninger’s sense, or a rigid-analytic or Berkovich structure place by place. Each changes what $\alpha \wedge d\alpha \neq 0$ means, and none is carried out here. This is the largest gap between the dictionary and a geometric statement, and it is separate from, and prior to, the positivity question of §6.

6. The Missing Rung: Global Positivity

6.1 Formulation

All the local machinery works. The remaining step is:

Conjecture 6.1 (Global Positivity / RH restated). *The natural action of the idele class group $\mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}$ on $\ker \alpha_{arith}$ is **positive-definite**. Equivalently, the total “twisting energy” of the arithmetic contact structure is strictly minimized precisely on the hyperplane $\sigma = \frac{1}{2}$, and any zero-curve at $\sigma \neq \frac{1}{2}$ would force $\alpha_{arith} \wedge d\alpha_{arith} = 0$ at some adelic point — a contradiction with maximal non-integrability.*

This conjecture is exactly the Riemann Hypothesis, restated in contact-arithmetic language. It is not easier to prove in this language; the arithmetic difficulty is fully preserved.

6.2 Why the reformulation is still useful

A clean reformulation can: - Suggest new proof strategies by connecting to known positivity results. - Reveal structural analogies with cases where RH is already proved (see Section 7). - Provide a unified geometric language for comparing disparate approaches (Connes, Weil, random matrix theory).

7. Comparison with Known Approaches

7.1 Weil’s explicit formula and positivity criterion

Weil (1952) showed that RH for ζ is equivalent to the **positivity of a certain distribution** on the adèle class space: for a suitable class of test functions h ,

$$\sum_{\rho} \hat{h}(\rho) \geq 0,$$

where the sum is over non-trivial zeros. This is a spectral positivity statement. In our language: Weil’s positivity is the statement that the idele class group action has non-negative spectrum, which is the L^2 analogue of our contact-geometric positive-definiteness condition. The two conditions are morally identical but technically formulated in different categories (measure theory vs. differential geometry).

Partial results. Zero-density estimates (e.g., $N(\sigma, T) \ll T^{A(1-\sigma)} \log^B T$ for $\sigma > \frac{1}{2}$) give unconditional bounds showing that “most” zeros satisfy the contact condition. These correspond to partial positivity of the arithmetic contact form on large subsets of the adèle class space.

7.2 Connes’ spectral triple

Connes (1999) proposes to realize the zeros of ζ as the **missing part of the spectrum** of an operator on $L^2(\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times})$. The absorption spectrum of a suitable Hamiltonian H would contain the zeros, and RH would follow if H were shown to be self-adjoint (Hermitian), forcing real eigenvalues — hence $\text{Im}(\rho) \in \mathbb{R}$, i.e., $\text{Re}(\rho) = \frac{1}{2}$.

Translation to contact language. The self-adjointness of H corresponds to the positive-definiteness of the idele-class action in our framework. Connes’ “Reeb vector field” is our contact-geometric Reeb flow. The spectral gap condition in Connes corresponds to the strict non-vanishing of $\alpha \wedge d\alpha$ away from $\sigma = \frac{1}{2}$.

The difference: Connes works in L^2 (spectral theory), we work on contact manifolds (differential geometry). The obstacle is the same: proving positivity.

7.3 Function-field analogue (where RH is proved)

Over a finite field $\mathbb{F}_q$, the analogue of $\zeta(s)$ is the **zeta function of a curve** $C/\mathbb{F}_q$, and the analogue of RH was proved by Weil (curves, 1948) and Deligne (varieties, 1974).

Contact form in the function-field case. The Euler product is finite in each degree, and the “frequencies” $\log p$ become discrete valuations $v(p)$ in the function field. The key simplifications:

1. The adèle ring $\mathbb{A}_k$ (for function field k) is *locally compact* and the relevant spaces are *finite-dimensional* in the relevant cohomological sense.
2. The positivity (Weil’s positivity for curves) follows from the **Riemann–Roch theorem**, which provides an explicit algebraic control over the zero distribution.
3. In contact terms: the “arithmetic contact form” over $\mathbb{F}_q$ has a **finite-dimensional Reeb flow**, and positive-definiteness can be verified by an explicit computation over the Jacobian of C .

Lesson for the number-field case. The function-field proof works because algebraic geometry (Riemann–Roch, Lefschetz trace formula) provides the positivity “for free.” Over $\mathbb{Q}$, there is no analogue of Riemann–Roch that directly controls the infinite-dimensional adèle class space. The contact form in Section 4 encodes all the same arithmetic, but the positivity argument must come from elsewhere — perhaps from a new analytic tool, a motivic cohomology result, or a non-commutative geometry calculation.

7.4 Other geometric and dynamical programmes

Every citation below was checked against the published record before inclusion; three in-circulation attributions were found wrong and are corrected here.

Deninger’s foliated dynamical systems. Deninger [11] proposes an infinite-dimensional foliated leaf space carrying an $\mathbb{R}$ -action whose closed orbits correspond to primes and whose transfer-operator trace reproduces the local factors. *Where it stops:* the space itself has never been constructed; the differential operators on it remain hypothetical. This is the closest programme in spirit to the present paper, and it fails at the same point — §5.5 — namely producing an actual geometric object rather than a dictionary.

Meyer’s spectral interpretation. Meyer [12] reformulates the explicit formula through a representation of the idele class group on a Hilbert space extension. *Where it stops:* self-adjointness of the generator is structurally equivalent to Weil positivity again.

The Bost–Connes system. Bost and Connes [13] encode the primes in the KMS states of a Hecke C^* -algebra, with a phase transition at inverse temperature $\beta = 1$. *Where it stops:* the system makes no contact with the critical strip, where the zeros are.

Morishita’s bridge. Morishita [14] constructs a continuous, Galois-equivariant and flow-anti-equivariant map from Deninger’s Witt-space foliated dynamical systems to Connes–Consani’s adelic spaces, carrying closed orbits to prime places. This is the most direct link between the two programmes the present construction sits between, and any serious attempt at §5.5 should start there rather than here.

The lift is a standard device, and should be recognised as one. Promoting a two-dimensional state space to a three-dimensional contact manifold carrying a 1-form of the shape $c dU - g dV$ is exactly the construction used in the contact geometry of thermodynamics, where the Gibbs form on the extended state space plays this role and the equations of state appear as Legendre submanifolds [19]. The present paper does not introduce that geometry; it applies it, with (c, g) the real and imaginary parts of $-\zeta'/\zeta$ in place of the conjugate thermodynamic pairs.

And the structure group is the right one. Mrugała [20] proves that for a thermodynamic system with n degrees of freedom, the structure group of the tangent bundle of the $(2n + 1)$ -dimensional thermodynamic phase space reduces to $U(n) \times 1$. The construction of §3 is the case $n = 1$: $\mathbb{R}_{(U,V)}^2 \times \mathbb{R}_t$ is $2 \cdot 1 + 1$ dimensional, and the reduced group is $U(1) \times 1$ — a circle acting on the contact plane, trivial along the remaining direction.

That is precisely the geometry this paper uses, and it was not chosen for that reason. The phase of ζ rotates in the (U, V) plane as t advances, and t itself is inert: the helices of Figure 1 are orbits of exactly this $U(1)$, and the $\times 1$ is the height coordinate they climb. It also explains why the form has *two* coefficients rather than one. $U(1)$ acts on a complex quantity, and (c, g) are not two independent functions but the real and imaginary parts of a single meromorphic object, $-\zeta'/\zeta$. The two-coefficient shape of $\alpha_{\text{arith}} = c d\tilde{U} - g d\tilde{V}$ is what the structure group requires, not a modelling choice — which is a better reason for it than the one given in §4.1.

The reading remains live rather than historical. Pernambuco and Céleri [21], working on quantum thermodynamics rather than on arithmetic, arrive at a thermodynamic gauge group $\mathcal{G}_T \simeq U(n_1) \times \cdots \times U(n_k)$ acting on degenerate energy subspaces, and identify irreversibility as a consequence of *restricted observability* under gauge reduction rather than of the underlying dynamics, which is reversible. Nothing in that work bears on ζ or on the claims of this paper, and it is cited only for the convergence: two independent routes into the geometry of thermodynamics, three decades apart, both landing on unitary groups as its symmetry.

We think this makes the contribution clearer rather than smaller. The transferable object is not a new kind of manifold but a **dictionary entry**: the assertion that the arithmetic of ζ can be written in the same extended-phase-space language, with the functional equation appearing as a reflection law on the coefficients (§4.5) and the zeros as punctures of the $U = V = 0$ axis (§4.4).

What is not claimed. No novelty is asserted for the lift, for the two-coefficient form, or for the one-sided pole of §4.4. Establishing which of these are unremarked would require a literature search we have not run, and an unsearched novelty claim is a claim of ignorance dressed as a result. What is claimed is exactly what §4.6 lists, and nothing else.

8. Structure of a Putative Proof

For completeness, we state what a proof of RH within this framework would require:

Step 1 (done): Construct α_{arith} and verify $\alpha_{\text{arith}} \wedge d\alpha_{\text{arith}} \neq 0$ on the complement of the zeros (Section 4–5).

Step 2 (done): Decompose into local forms α_v and verify the valuation lock at each p -adic place (Section 5.3).

Step 3 (open): Prove that the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ implies a **geometric symmetry** $\Phi : \sigma \mapsto 1 - \sigma$ of the contact structure, pairing α_{arith} at σ with its dual at $1 - \sigma$.

Step 4 (open, core difficulty): Prove that Φ -symmetry together with maximal non-integrability forces every zero-curve to lie on the fixed locus $\sigma = \frac{1}{2}$ of Φ — i.e., prove that any zero at $\sigma_0 \neq \frac{1}{2}$ would force $\alpha_{\text{arith}} \wedge d\alpha_{\text{arith}} = 0$ at some adelic point.

Step 4 is precisely Global Positivity (Conjecture 6.1) and is equivalent to RH.

9. Discussion and Open Questions

1. **Is there a Morse-theoretic approach to Step 4?** The “twisting energy” minimization language suggests a Morse-theory or symplectic-filling argument might work, analogous to how filling obstructions appear in 3-manifold contact topology (Giroux, Etnyre).
2. **Can the DNLS nonlinearity be tuned to approximate the prime distribution?** The reduced DNLS contact form (Section 2.4) produces chaotic helices whose statistics can be adjusted via β and γ . Is there a parameter regime where the local zero-spacing statistics match GUE (the random matrix prediction for zeta zeros)?
3. **Adelic contact geometry as a field.** The construction of α_{arith} on $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ is, to our knowledge, new. Does this space admit a well-defined contact topology (tight vs. overtwisted contact structures)? If α_{arith} defines a *tight* contact structure, that would be a strong rigidity result consistent with RH.
4. **p -adic Reeb dynamics.** The local Reeb vector field at each place p is the dual vector field to α_p (defined by $\alpha_p(R_p) = 1$, $d\alpha_p(R_p, \cdot) = 0$). Studying the dynamics of R_p in the rigid-analytic setting might reveal additional arithmetic structure.

Appendix A: Meromorphic Continuation of $g(\sigma, t)$ Inside the Critical Strip

For $\sigma > 1$, the series $g(\sigma, t) = \sum_n \Lambda(n)n^{-\sigma} \sin(t \log n)$ converges absolutely. For $0 < \sigma \leq 1$, the Dirichlet series for $-\zeta'/\zeta$ diverges, but ζ'/ζ extends meromorphically to $\mathbb{C}$ (with simple poles at zeros of ζ and a double pole at $s = 1$). We define $g(\sigma, t)$ inside the strip as

$$g_{\text{cont}}(\sigma, t) = -\text{Im} \left(-\frac{\zeta'}{\zeta}(\sigma + it) \right),$$

using this meromorphic continuation. The exterior derivative formula $d\alpha_{\text{arith}} = -(\partial_t g_{\text{cont}}) dt \wedge dU$ remains valid wherever $\zeta \neq 0$ and ζ' exists. At a zero s_0 , the logarithmic derivative has a simple pole, and $g_{\text{cont}} \rightarrow \pm\infty$: the contact planes twist infinitely rapidly, consistently with the zero-curve piercing $U = V = 0$.

Appendix B: Summary Table — Parallel Structures

Concept	Smooth ODE (Grossi Ch. 2)	Arithmetic (this paper)	Connes' approach
Phase space	$\mathbb{R}^2 \times \mathbb{R}_t$	$\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$	$L^2(\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times})$
Contact form	$\alpha = dy - g(x, y) dx$	$\alpha_{\text{arith}} = dV - g_{\text{cont}} dU$	Spectral triple (A, H, D)
Non-integrability	$\alpha \wedge d\alpha \neq 0$ (local, automatic)	$\alpha_{\text{arith}} \wedge d\alpha_{\text{arith}} \neq 0$ (global, non-trivial)	D self-adjoint
RH condition	N/A (smooth zeros are fine)	Global positivity of idele-class action	Self-adjointness of H
Forcing mechanism	Smooth ODE rigidity	$\mathbb{Q}$ -independence of $\{\log p\} + p$ -adic valuation locks	Trace formula positivity
Status	Proved (trivially)	Open (= RH)	Open (= RH)
Function-field analogue	Same framework, finite-dim	Weil positivity via Riemann–Roch	Proven (Deligne)

Deposit Contents and Reproducibility

Everything that produces a number or a figure in this paper is deposited beside it, and every claim in §4.6 resolves to one of these files.

File	Bytes	What it is
<code>rh_arithmetic_contact_v3.pdf</code>	49,185	this paper
<code>RH_arithmetic_contact_structures.md</code>	4,480	its source
<code>ZetaReflection.lean</code>	11,344	the formalisation of §4.7. Lean 4.32.0 / Mathlib v4.32.0
<code>RELATED-WORK.md</code>	7,012	the citation audit behind §7.4, including what was rejected
<code>verify_reflection_laws.py</code>	3,311	the 30-digit numerical checks of §4.5 and (FE-log)
<code>figures.py</code>	2,847	generates Figure 1
<code>fig1_rh_lift.pdf / .png</code>	43,129 / 278,509	Figure 1

Byte counts are given so that a reader can tell at a glance whether the file they downloaded is the one this paper describes. `ZetaReflection.lean` must be **11,344 bytes** and must contain `chiLog_real_on_critical_line`, `Zlog_conj`, `gCoef_odd_in_t` and `cCoef_even_in_t`. If it does not, the deposit is wrong and this table is how you find out.

Repositories. Source and Lean: [TOTOGT/GTCT](#), under `deposits/rh-arithmetic-contact-v1/`, with the formalisation also at `book4/ZetaReflection.lean`. Wider corpus: [TOTOGT/geometry](#) (*Principia Orthogona*, Books 1–8) and [TOTOGT/AXLE](#).

To check the formalisation yourself. In any Lean project pinned to Mathlib v4.32.0:

```
lake env lean ZetaReflection.lean
```

then append `#print axioms Book4.Ch12.<name>` for each declaration. A claim in §4.6 marked *machine-checked* must report a subset of `[propext, Classical.choice, Quot.sound]`. If it reports `sorryAx`, the statement is admitted, not proved; if it reports a `native_decide` axiom, it is compiler-trusted, not kernel-checked. Both are reasons to disbelieve the table, and both have been found in this corpus before.

To check the numerics. `python3 verify_reflection_laws.py` (requires `mpmath`) prints the residuals for the reflection laws and for (FE-log) at the sample points, at 30 significant digits.

References

Entries marked **[verified]** were verified against the published record while preparing v2. Where a commonly circulated citation is wrong, the correction is noted.

1. B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte der Berliner Akademie (1859).
2. A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund (1952), 252–265.
3. [verified] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.
4. A. Connes and C. Consani, *On the notion of geometry over $\mathbb{F}_1$* , J. Algebraic Geom. **20** (2011), 525–557.
5. H. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. **24** (1973), 181–193.
6. P. N. Grossi, *Principia Orthogona*, Book 4, Chapter 2: Contact geometry and the extended phase space. <https://totogt.github.io/geometry/book4/ch02.html>
7. H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications **53** (2004).
8. V. I. Arnold, *Mathematical Methods of Classical Mechanics*, Springer (1989).
9. J. Etnyre, *Introductory lectures on contact geometry*, Proc. Sympos. Pure Math. **71** (2003).
10. P. Deligne, *La conjecture de Weil I*, Publ. Math. IHÉS **43** (1974), 273–307.
11. [verified] C. Deninger, *Some analogies between number theory and dynamical systems on foliated spaces*, Documenta Math., Extra Vol. ICM I (1998), 163–186. (*Frequently miscited as “...and topology”, and to Ergod. Th. & Dynam. Sys.; both are wrong.*)
12. [verified] R. Meyer, *On a representation of the idele class group related to primes and zeros of L-functions*, Duke Math. J. **127** (2005); preprint arXiv:math/0311468. (*Often truncated to “...related to prime numbers”.*)
13. [verified] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. (N.S.) **1** (1995), 411–457. (*Sometimes miscited as “...and arithmetical dynamical systems”.*)
14. [verified] M. Morishita, *On a relation between Deninger’s foliated dynamical systems and Connes–Consani’s adelic spaces*, arXiv:2508.15971 (2025).
15. [verified] A. Connes, C. Consani and M. Marcolli, *The Weil proof and the geometry of the adèles class space*, in *Algebra, Arithmetic and Geometry* (Manin Festschrift), Progr. Math.

- 269**, Birkhäuser (2009).
16. E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., rev. D. R. Heath-Brown, Oxford University Press (1986). §3.3 (local expansions near zeros), §6.5 (phase of $Z(t)$).
 17. H. Davenport, *Multiplicative Number Theory*, 3rd ed., rev. H. L. Montgomery, Springer GTM **74** (2000).
 18. The mathlib Community, *The Lean Mathematical Library*, CPP 2020. Version v4.32.0 used throughout.
 19. [verified] R. Mrugała, *Continuous contact transformations in thermodynamics*, Rep. Math. Phys. **33** (1993), 149–154. (*Circulated with the wrong title, year, volume and pages — “...in thermodynamic phase space”, 1990, 29(1), 117–129 — none of which match the record.*) For the underlying contact structure on the extended thermodynamic state space see also R. Hermann, *Geometry, Physics and Systems*, Marcel Dekker (1973).
 20. [verified] R. Mrugała, *Structure group $U(n) \times 1$ in thermodynamics*, J. Phys. A: Math. Gen. **38** (2005), no. 50, 10905–10916. doi:10.1088/0305-4470/38/50/003. Verified against Crossref, 30 August 2026.
 21. T. Pernambuco and L. C. Céleri, *Geometry of restricted information: the case of quantum thermodynamics*, arXiv:2602.06716 [quant-ph] (2026). Cited for convergence only; it makes no claim bearing on this paper.